

WORK SHEET - 08

S-BLOCK & HYDROGEN COMPOUND & WEAK FORCES

Comprehension

Question No. 1 to 3 (3 questions)

The existence of intermolecular forces is supported by the facts : non ideality of real gases, **Joule-Thomson effect**, liquefaction of gases. The electrical field of a dipole can induce a dipole moment in adjacent molecule (which may be polar or non polar) then the induced dipole can interact electrostatically with the polarising dipole.

- Q.1 The strongest force among the following is
(A) London force (B) Ion-dipole interaction
(C) Dipole-induced dipole interaction (D) Dipole-dipole interaction
- Q.2 Select the **correct** statement :
(A) Boiling point of NF_3 is greater than NMe_3 .
(B) Greater the dipole moments, then greater will be the dipole-dipole interaction.
(C) London dispersion force increases with decreasing number of electrons.
(D) Boiling point of hydrides of carbon family decreases down the group.
- Q.3 Noble gases can be liquified due to " _____ " within molecules :
(A) ion-dipole interaction
(B) dipole - dipole interaction
(C) dipole - induced dipole interaction
(D) instantaneous dipole - induced dipole interaction

Question No. 4 to 6 (3 Question)

All the anions are detected from soda extract except CO_3^{2-} and HCO_3^- . It is very useful when any water insoluble salt is present in the given mixture.

- Q.4 For which of the following salt preparation of soda extract is essential.
(A) $(\text{NH}_4)_2\text{SO}_4$ (B) Na_2S (C) KBr (D) AgCl
- Q.5 Which of the following salt remains unchanged during preparation of soda extract.
(A) BaS_2O_3 (B) PbSO_4 (C) CaCO_3 (D) Na_2S
- Q.6 Choose the **correct** statement regarding soda extract preparation.
(A) Instead of Na_2CO_3 , Na_2SO_4 can also be used
(B) All Na-salts are usually soluble in water
(C) K_2CO_3 also can be used instead of Na_2CO_3 .
(D) During soda extract preparation Hg_2Cl_2 gives metallic Hg as residue.

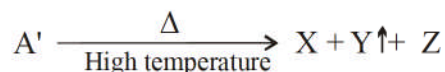
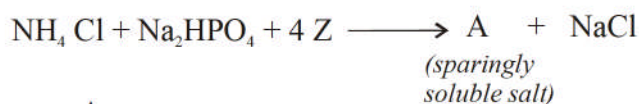
Question No. 7 to 9 (3 Question)

Quick lime $\xrightarrow{\text{H}_2\text{O}}$ (A) $\xrightarrow[\text{Passed}]{\text{Cl}_2 \text{ is}}$ (B) + H_2O
(dried)

- Q.7 (B) on reaction with NH_3 & PH_3 produces _____ & _____ respectively.
(A) N_2 & P_4 (B) NCl_3 & PCl_3 (C) N_2 & PCl_3 (D) NCl_3 & PCl_5

- Q.8 The oxidation states of Cl-atoms in B are
 (A) +1, +1 (B) +1, 0 (C) -1, +1 (D) -1, -1
- Q.9 When aqueous solution of (A) reacts with gas mixture (D) and turns milky. (D) may consist of
 (A) $\text{CO}_2 + \text{H}_2$ (B) $\text{SO}_2 + \text{N}_2$ (C) $\text{CO}_2 + \text{SO}_2$ (D) All of the three

Question No. 10 to 12 (3 Question)

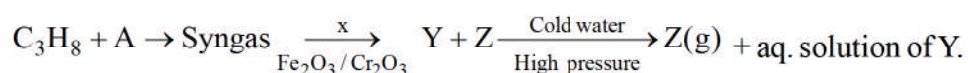


Compound (Z) is a neutral oxide and turns anhydrous CuSO_4 blue. Gas (Y) is a basic gas which can also be obtained by treating ammonium sulphate with lime water. Sequences mentioned above involve balanced reactions.

- Q.10 The compound 'X' is
 (A) NaHPO_4 (B) Na_3PO_3 (C) NaPO_3 (D) Na_3PO_4
- Q.11 Which of the following reactions would produce gas 'Y'?
 (A) $\text{Mg}_3\text{N}_2 + \text{H}_2\text{O} \longrightarrow$ (B) $\text{Zn} + \text{NaOH} + \text{NaNO}_3 \xrightarrow{\Delta}$
 (C) $(\text{NH}_4)_2\text{CO}_3 \xrightarrow{\Delta}$ (D) All of these
- Q.12 Formula of (A) is :
 (A) $\text{Na}_2(\text{NH}_4)\text{PO}_4 \cdot 4\text{H}_2\text{O}$ (B) $\text{Na}(\text{NH}_4)\text{HPO}_4 \cdot 4\text{H}_2\text{O}$
 (C) $\text{Na}(\text{NH}_4)_2\text{PO}_4 \cdot 4\text{H}_2\text{O}$ (D) $(\text{NH}_4)_2\text{HPO}_4 \cdot 4\text{H}_2\text{O}$

Question No. 13 to 16 (4 Question)

Consider the following flowsheet and answer the question given after it



Z has low chemical reactivity at room temperature but under vigorous suitable conditions it reacts with other elements to form very useful compounds. Z is also being looked upon as prospective source of energy for future.

- Q.13 The above flow sheet refers to which process-
 (A) Lane's process (B) Haber's process (C) Bosch process (D) Mond's process
- Q.14 Which two substances are same ?
 (A) X and Z (B) A and X (C) A and Y (D) A and Z
- Q.15 Which of following represents dihydrogen ?
 (A) Y (B) Z (C) X (D) None of these
- Q.16 Components of syn gas are-
 (A) CO_2 , CO (B) CO, H_2O (C) CO, H_2 (D) CO_2 , H_2

Single correct

- Q.17 The **correct** order of solubility in water, of the sulphates of group II elements would be :
(A) $\text{BeSO}_4 < \text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4 < \text{BaSO}_4$
(B) $\text{MgSO}_4 < \text{CaSO}_4 < \text{SrSO}_4 < \text{BaSO}_4 < \text{BeSO}_4$
(C) $\text{BeSO}_4 > \text{MgSO}_4 > \text{CaSO}_4 > \text{SrSO}_4 > \text{BaSO}_4$
(D) $\text{MgSO}_4 > \text{BeSO}_4 > \text{CaSO}_4 > \text{SrSO}_4 > \text{BaSO}_4$
- Q.18 The reagent commonly used to determine & remove hardness of water is-
(A) oxalic acid (B) disodium salt of EDTA
(C) sodium citrate (D) sodium thiosulphate
- Q.19 Calcium imide on hydrolysis will give gas (B) which on oxidation by bleaching powder gives gas (C). Gas (C) on reaction with magnesium give compound (D). (D) on hydrolysis gives again gas (B). (B), (C) and (D) are
(A) NH_3 , N_2 , Mg_3N_2 (B) N_2 , NH_3 , MgNH
(C) N_2 , N_2O_5 , $\text{Mg}(\text{NO}_3)_2$ (D) NH_3 , NO_2 , $\text{Mg}(\text{NO}_2)_2$
- Q.20 The pair of compounds which cannot exist together in aqueous solution is:
(I) NaH_2PO_4 and NaHCO_3 (II) Na_2CO_3 and NaHCO_3
(III) NaOH and NaH_2PO_4 (IV) NaHCO_3 and NaOH
(A) I, II, III (B) II, III (C) I, IV (D) only IV
- Q.21 Which one of the following reactions is not associated with the solvay process of manufacture of sodium carbonate-
(A) $\text{NH}_3 + \text{H}_2\text{O} + \text{CO}_2 \rightarrow \text{NH}_4\text{HCO}_3$
(B) $\text{NaCl} + \text{NH}_4\text{HCO}_3 \rightarrow \text{NaHCO}_3 + \text{NH}_4\text{Cl}$
(C) $2\text{NaHCO}_3 \xrightarrow{\Delta} \text{Na}_2\text{CO}_3 + \text{H}_2\text{O} + \text{CO}_2$
(D) $2\text{NaOH} + \text{CO}_2 \rightarrow \text{Na}_2\text{CO}_3 + \text{H}_2\text{O}$
- Q.22 (i) $\text{A} + \text{Na}_2\text{CO}_3 \longrightarrow \text{B} + \text{C}$
(ii) $\text{A} \xrightarrow{\text{CO}_2} (\text{Milky}) \text{C}$
The chemical formula of A and B are respectively
(A) NaOH & $\text{Ca}(\text{OH})_2$ (B) $\text{Ca}(\text{OH})_2$ and NaOH
(C) NaOH and CaO (D) CaO & $\text{Ca}(\text{OH})_2$
- Q.23 Which of the following carbide does not release any hydrocarbon on reaction with water?
(A) SiC (B) Be_2C (C) CaC_2 (D) Mg_2C_3
- Q.24 The correct order of thermal stability of the given azides is
(A) $\text{AgN}_3 > \text{AuN}_3 > \text{CuN}_3$ (B) $\text{AgN}_3 < \text{AuN}_3 < \text{CuN}_3$
(C) $\text{AuN}_3 < \text{AgN}_3 < \text{CuN}_3$ (D) $\text{AuN}_3 = \text{AgN}_3 = \text{CuN}_3$
- Q.25 H_2O_2 restores the colour of old oily paintings, blackened by the action of H_2S gas by-
(A) Converting PbO_2 to Pb (B) By oxidising PbS to PbSO_4
(C) Converting PbCO_3 to Pb (D) Oxidising PbSO_3 to PbSO_4

- Q.26 Choose the **correct** statement(s).
- (I) BeCO_3 is kept in the atmosphere of CO_2 since it is least thermally stable.
 (II) Be dissolves in an alkali forming $[\text{Be}(\text{OH})_4]^{2-}$
 (III) BeF_2 forms complex ion with NaF in which Be goes with cation.
 (IV) BeF_2 forms complex ion with NaF in which Be goes with anion.
 (A) I, II, III (B) II & III (C) I, II, III & IV (D) I, II & IV
- Q.27 CaO is an ionic compound, not a covalent compound because:
 (A) due to high ionisation energy of calcium
 (B) due to high electron affinity of oxygen
 (C) due to high lattice energy of Ca^{+2} and O^{-2}
 (D) due to high ionisation energy and low electron affinity of calcium and oxygen respectively.
- Q.28 Which of the following option is **correct** regarding Li^+ and Be^{2+} ?
 (A) Size of $\text{Li}^+ < \text{Be}^{2+}$
 (B) Hydration energy of Li^+ and Be^{2+} are equal
 (C) Li^+ and Be^{2+} having same molar conductivity in aqueous solution
 (D) Size of $\text{Li}^+ > \text{Be}^{2+}$
- Q.29 Ionic hydrides react with water to give-
 (A) Acidic solutions (B) Basic solutions (C) Hydride ion (D) Protons
- Q.30 Which of the following is not a water softener?
 (A) Na_2CO_3 (B) Na_2SO_4 (C) Calgon (D) Permutit
- Q.31 The catalyst used in process of manufacture of H_2 from water gas is-
 (A) Finely divided Ni (B) V_2O_5 (C) Pb (D) $\text{Fe}_2\text{O}_3 + \text{Cr}_2\text{O}_3$
- Q.32 Which of the following is not true?
 (A) Hardness of water is shown by its behaviour towards soap
 (B) The temporary hardness is due to the presence of Ca and Mg bicarbonates
 (C) Permanent hardness is due to the presence of soluble Ca and Mg sulphates and chloride
 (D) Permanent hardness can be removed by boiling the water
- Q.33 Which statement is wrong-
 (A) Ordinary hydrogen is an equilibrium mixture of ortho and para hydrogen
 (B) In ortho hydrogen spin of two nuclei is in same direction
 (C) Ortho and para forms do not resemble in their chemical properties
 (D) In para hydrogen spin of two nuclei is in opposite direction
- Q.34 Hydrogen can be prepared by mixing steam and water gas at 500°C in the presence of Fe_2O_3 and Cr_2O_3 . This process is called-
 (A) Lane process (B) Hydrolith Method (C) Bosch process (D) None of these
- Q.35 Which of the following method is used for preparation of hydrogen?
 (A) Uyneo's method (B) Dennis method (C) Deacon's process (D) Down cell method
- Q.36 Which of the following has maximum tendency to form nitride with N_2 ?
 (A) Na (B) Li (C) K (D) Cs

- Q.37 Very pure hydrogen (99.9%) can be made by which of the following processes ?
 (A) mixing natural hydrocarbons of high molecular weight
 (B) electrolysis of water
 (C) reaction of salt like hydrides with water
 (D) reaction of methane with steam
- Q.38 CO_2 gas along with solid (Y) is obtained when sodium salt (X) is heated. (X) is again obtained when CO_2 gas is passed into (Y). X & Y are-
 (A) Na_2CO_3 , Na_2O (B) Na_2CO_3 , NaOH
 (C) NaHCO_3 , Na_2CO_3 (D) Na_2CO_3 , NaHCO_3
- Q.39 Common table salt becomes moist and does not pour easily in rainy seasons because-
 (A) it contains magnesium chloride (B) it contains magnesium carbonate
 (C) it melts slightly in rainy season (D) sodium chloride is hygroscopic
- Q.40 BeF_2 is soluble in water, whereas, the fluorides of other alkaline earth metals are insoluble because of-
 (A) ionic nature of BeF_2 (B) greater hydration energy of Be^{2+} ion
 (C) covalent nature of BeF_2 (D) none of the above
- Q.41 Permanent hardness of water can be removed by adding calgon (NaPO_3)_n. This is an example of-
 (A) Adsorption (B) Exchange of ion
 (C) Precipitation (D) None of these
- Q.42 Which salt doesn't cause hardness of water?
 (A) $\text{Mg}(\text{HCO}_3)_2$ (B) CaCl_2 (C) MgSO_4 (D) MgCO_3
- Q.43 Sodium metal is highly reactive and cannot be stored under
 (A) Toluene (B) Kerosene oil (C) Alcohol (D) Benzene
- Q.44 The compound of calcium (X) is used in sugar industry for the purification of sugar. When exposed to any oxy-hydrogen flame, it becomes incandescent and starts emitting white light. On treatment with CO_2 it forms a compound, which can be decomposed to give back X at very high temperature X is-
 (A) CaCO_3 (B) CaO (C) $\text{Ca}(\text{OH})_2$ (D) CaSO_4
- Q.45 Which of the following is wrong-
 (A) Reducing character of alkaline earth metals increases from Be to Ba
 (B) $\text{Be}(\text{OH})_2$ is amphoteric in nature
 (C) The solubilities of sulphates and carbonates decrease with increase in atomic number of alkaline earth metals
 (D) BeCl_2 has much higher mp and insoluble in organic solvents
- Q.46 When hydrolith is treated with water, it yields.
 (A) H_2 (B) H_2O_2 (C) N_2 (D) NaH
- Q.47 Hydrogen peroxide is manufactured by the autoxidation of-
 (A) 2-ethylanthraquinol (B) Anthraquinone
 (C) Naphthalene (D) Anthracene
- Q.48 In Bosch's process which gas is utilised for the production of hydrogen-
 (A) Producer gas (B) Water gas (C) Coal gas (D) Natural gas

Assertion Reason

Q.49 **Statement-1:** Superoxides of alkali metals are paramagnetic.

Statement-2: Superoxides contain the ion O_2^- which has one unpaired electron in its anti-bonding molecular orbital.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q.50 **Statement-1:** Alums are isomorphous crystalline double salts, which are soluble in water.

Statement-2: The aq. solutions of alums have acidic character due to hydrolysis.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q.51 **Statement-1:** The hydrogen carbonates of the alkali metals are soluble in water, but are less soluble than the corresponding normal carbonates.

Statement-2: The hydrogen carbonates of the alkali metals have H-bonding.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

Q.52 **Statement-1:** Temporary hardness can be removed by boiling.

Statement-2: Bicarbonates are precipitated in form of carbonates due to boiling.

(A) Statement-1 is true, statement-2 is true and statement-2 is correct explanation for statement-1.

(B) Statement-1 is true, statement-2 is true and statement-2 is NOT the correct explanation for statement-1.

(C) Statement-1 is true, statement-2 is false.

(D) Statement-1 is false, statement-2 is true.

More than one may be correct

Q.53 Which statement(s) is / are **correct** regarding alkali metals?

(A) Thermal stability of metal nitrate increases on moving upward the group.

(B) Water solubility of metal carbonate increases on moving down the group.

(C) All metal bicarbonate salts exist in solution form except $NaHCO_3$.

(D) B.P. and M.P. decreases down the group.

Q.54 Which of the following will not give any colour to flame?

(A) Be

(B) Mg

(C) Na

(D) Li

Q.55 Carbon disulphide reacts in aq. NaOH solution producing.

(A) Na_2S

(B) Na_2CO_3

(C) Na_2SO_3

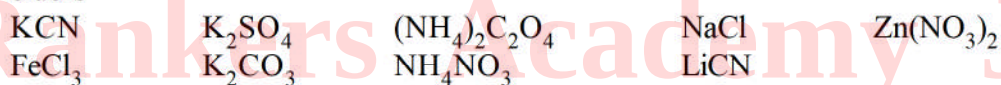
(D) Na_2CS_3

Match the column

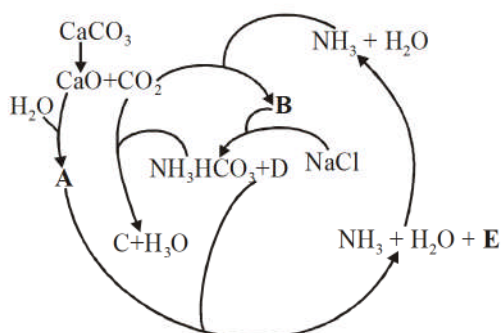
Q.56	Column-I	Column-II
(A)	Increasing order of solubility	(P) LiOH, KOH, CsOH
(B)	Increasing order of thermal stability	(Q) Li_2CO_3 , Na_2CO_3 , K_2CO_3
(C)	Decreasing order of covalent character	(R) CaC_2O_4 , SrC_2O_4 , BaC_2O_4
		(S) MgO, CaO, BaO
Q.57	Column - I (Name of the compound)	Column - II (Formula)
(A)	Dead burnt plaster	(P) CaCl_2
(B)	Quicklime	(Q) $\text{Ca}(\text{OH})_2$
(C)	Slaked lime	(R) CaO
(D)	Gypsum	(S) CaSO_4
		(T) $\text{CaSO}_4 \cdot 2\text{H}_2\text{O}$

Subjective :

Q.58 Amongst the following, the total number of compounds whose aqueous solution turns red litmus paper blue is

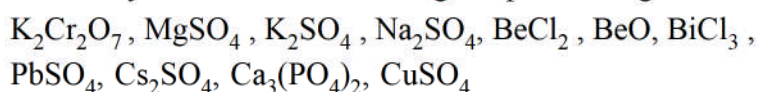


Q.59 The Haber-process can be represented by the following scheme.



Identify A, B, C, D and E.

Q.60 How many total number of following compound will give flame colour test in Bunsen flame?



ANSWER KEY

Q.1	B	Q.2	B	Q.3	D	Q.4	D	Q.5	C
Q.6	B	Q.7	C	Q.8	C	Q.9	D	Q.10	C
Q.11	D	Q.12	B	Q.13	C	Q.14	B	Q.15	B
Q.16	C	Q.17	C	Q.18	B	Q.19	A	Q.20	D
Q.21	D	Q.22	B	Q.23	A	Q.24	C	Q.25	
Q.26	D	Q.27	C	Q.28	D	Q.29	B	Q.30	B
Q.31	D	Q.32	D	Q.33	C	Q.34	C	Q.35	A
Q.36	B	Q.37	C	Q.38	C	Q.39	A	Q.40	B
Q.41	B	Q.42	D	Q.43	C	Q.44	B	Q.45	D
Q.46	A	Q.47	A	Q.48	B	Q.49	A	Q.50	B
Q.51	A	Q.52	A	Q.53	BCD	Q.54	AB	Q.55	BD
Q.56	(A) PQRS, (B) PQR (C) PQRS			Q.57	(A) S (B) R (C) Q (D) T			Q.58	3
Q.59		Q.60	8						

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